STAT 526 HW 8

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Problem 0

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Problem 1

> The log-log confidence interval for S(t) is often recommended. Given that $Var(log[S(t)]) \approx \sum \frac{d_j}{(n_j - d_j)n_j}$ use the delta method to approximate the variance of log(-log[S(t)]). Make sure to show all the calculation steps.

We start with our random variable being $Y = \log[\hat{S}(t)]$ and then are transforming it with log, ie finding $\log(-\log[\hat{S}(t)])$. For the delta method, we know $\text{Var}(g(Y)) \approx [g'(\mu_Y)]^2 \text{Var}(Y)$.

Then $g(Y) = \log(-Y)$.

We calculate the first derivative:

$$g'(Y) = \frac{d}{dY}\log(-Y) = \frac{1}{-Y} \cdot (-1) = \frac{1}{Y}$$

Which means:

$$g'(\log[\hat{S}(t)]) = \frac{1}{\log[\hat{S}(t)]}$$

Now we plug into our delta method equation:

$$\widehat{\text{Var}}(\log(-\log[\hat{S}(t)])) \approx \left[\frac{1}{\log[\hat{S}(t)]}\right]^2 \widehat{\text{Var}}(\log[\hat{S}(t)]) = \frac{1}{[\log \hat{S}(t)]^2} \sum_{t_j \leq t} \frac{d_j}{(n_j - d_j)n_j}$$

Problem 2

The survival times of 11 patients with Acute Myelogenous Leukemia are given below: 9, 13, 13+, 18, 23, 28+, 31, 34, 45+, 48, 161+, where censored data are marked by "+" in the superscript.

a.)

Calculate the Kaplan-Meier estimate of S(20).

Our KM estimate:

$$\hat{S}(t) = \prod_{i:t_i \leq t} \left(1 - \frac{d_i}{n_i}\right)$$

t_i	n_i (At Risk)	d_i (Events)	$1 - d_i/n_i$	$\hat{S}(t_i)$
9	11	1	10/11	0.9091
13	10	1	9/10	0.8182
18	8	1	7/8	0.7159

Next death at t=23 so we do not need to calculate that far. Our final value is 0.7159 as our estimate for S(20).

b.)

Obtain the standard error of $log\hat{S}(20)$.

We use the identity given to us earlier.

$$Var(\log \hat{S}(t)) = \sum_{i:t_i \leq t} \frac{d_i}{n_i(n_i - d_i)}$$

Looking at our table earlier, we can grab these numbers for the time events 9,13,18. We have $t = 9$: $\frac{1}{11(10)} = 0.00909$, $t = 13$: $\frac{1}{10(9)} = 0.01111$, $t = 18$: $\frac{1}{8(7)} = 0.01786$.

Then $Var(\log \hat{S}(20)) = 0.00909 + 0.01111 + 0.01786 = 0.03806$ and $SE(\log \hat{S}(20)) = \sqrt{0.03806} \approx \mathbf{0.1951}$.

c.)

Assuming an exponential distribution, estimate S(20).

We know $S(t) = e^{-\lambda t}$. MLE for λ is $\hat{\lambda} = \frac{D(t)}{\sum T_i}$.

We can calculate $\hat{\lambda} = \frac{7}{423} \approx 0.01655$ and then $\hat{S}(20) = e^{-0.01655 \times 20} = e^{-0.331} \approx 0.7182$

d.)

Obtain the standard error of $log\hat{S}(20)$ for the exponential estimate

We know $\log S(t) = -\lambda t$ under exponential modeling.

If we assume it is exponential then variance of $\hat{\lambda}$ is $Var(\hat{\lambda}) = \frac{\lambda^2}{D}$.

Combine these and we can calculate the variance:

$$Var(\log \hat{S}(t)) = Var(-\hat{\lambda}t) = t^2 Var(\hat{\lambda}) = \frac{t^2\hat{\lambda}^2}{D}$$

Now we plug the values we got in part c in:

$$Var(\log \hat{S}(20)) = \frac{20^2 \times (0.01655)^2}{7} = \frac{400 \times 0.0002739}{7} \approx 0.01565$$

Take the square root and we get $SE(\log \hat{S}(20)) = \sqrt{0.01565} \approx 0.1251$

Problem 3

The following data report the survival of 30 late stage liver cancer patients (in weeks) according to three different treatment methods. In Homework #7, you tested for differences among the treatments nonparametrically using the log-rank test and parametrically assuming the survival times were exponential.

a.)

Test whether the exponential or Weibull distributions best fit the data.

Use libraries Prof Song uses.

```r
library(survival)
library(survminer)
```

```
## Warning: package 'survminer' was built under R version 4.5.3

## Loading required package: ggplot2

## Loading required package: ggpubr

## Warning: package 'ggpubr' was built under R version 4.5.3

##
## Attaching package: 'survminer'

## The following object is masked from 'package:survival':
##
##     myeloma
```

```r
# Create the dataset
time <- c(2, 2, 4, 6, 7, 9, 12, 17, 19, 24, # Trt 1
          1, 1, 3, 4, 6, 6, 8, 9, 10, 12, # Trt 2
          5, 6, 8, 10, 12, 13, 13, 14, 19, 21) # Trt 3

cens <- c(0, 1, 1, 0, 1, 1, 1, 0, 0, 1, # Trt 1
            1, 1, 1, 0, 1, 1, 0, 1, 1, 0, # Trt 2
            0, 1, 1, 1, 0, 1, 1, 0, 1, 1) # Trt 3

trt <- factor(rep(1:3, each = 10))
surv_obj <- Surv(time, cens)

fit_weibull <- survreg(surv_obj ~ trt, dist = "weibull")
fit_exponential <- survreg(surv_obj ~ trt, dist = "exponential")
summary(fit_weibull)
```

```
##
## Call:
## survreg(formula = surv_obj ~ trt, dist = "weibull")
```

```
##               Value Std. Error     z      p
## (Intercept)  2.7741     0.2433 11.40 <2e-16
## trt2        -0.6577     0.3316 -1.98 0.0473
## trt3        -0.0151     0.3320 -0.05 0.9638
## Log(scale)  -0.5175     0.1853 -2.79 0.0052
##
## Scale= 0.596
##
## Weibull distribution
## Loglik(model)= -68.8   Loglik(intercept only)= -71
##  Chisq= 4.24 on 2 degrees of freedom, p= 0.12
## Number of Newton-Raphson Iterations: 7
## n= 30
```

```r
summary(fit_exponential)
```

```
##
## Call:
## survreg(formula = surv_obj ~ trt, dist = "exponential")
##               Value Std. Error     z       p
## (Intercept)  2.8332     0.4082  6.94 3.9e-12
## trt2        -0.6848     0.5563 -1.23    0.22
## trt3         0.0167     0.5563  0.03    0.98
##
## Scale fixed at 1
##
## Exponential distribution
## Loglik(model)= -72   Loglik(intercept only)= -73
##  Chisq= 2.01 on 2 degrees of freedom, p= 0.37
## Number of Newton-Raphson Iterations: 4
## n= 30
```

Use simple likelihood ratio test to test if exponentional model (special version of weibull) is better.

```r
L_weibull <- logLik(fit_weibull)
L_exp <- logLik(fit_exponential)
LR_stat <- as.numeric(2 * (L_weibull - L_exp))
pchisq(LR_stat, df = 1, lower.tail = FALSE)
```

```
## [1] 0.012053
```

Since the p-value is less than $\alpha = 0.05$, we reject the exponential model in favor of the weibull model.

b.)

> Assuming the survival times are Weibull, what are the mean survival lives for each group?

We know that the expected value of Weibull distribution is $E[T] = \lambda\Gamma\left(1 + \frac{1}{k}\right)$ with $k = 1/\text{scale}$ (from the survreg output) and $\lambda = \exp(\text{intercept} + \beta_i)$.

We get these values from our models above.

```r
coeffs <- coef(fit_weibull)
shape_k <- 1 / fit_weibull$scale

# Calculate means for each treatment group
# Trt 1 (intercept/baseline)
mean_trt1 <- exp(coeffs[1]) * gamma(1 + 1/shape_k)

# Trt 2
mean_trt2 <- exp(coeffs[1] + coeffs[2]) * gamma(1 + 1/shape_k)

# Trt 3
mean_trt3 <- exp(coeffs[1] + coeffs[3]) * gamma(1 + 1/shape_k)

mean_trt1
```

```
## (Intercept)
##    14.31066
```

```r
mean_trt2
```

```
## (Intercept)
##    7.413575
```

```r
mean_trt3
```

```
## (Intercept)
##    14.09647
```

Our mean survival times are 14.31066 for treatment 1, 7.413575 for treatment 2, 14.09647 for treatment 3.

c.)

> The Cox proportional hazard model was fit giving the following output. The estimates of survival functions for the first treatment group are $\hat{S}(1) = 0.9612$ and $\hat{S}(2) = 0.9405$. Compute the estimates of the survival functions of the corresponding values for the second and third groups, respectively.

Treatment group 1 is the baseline (intercept) and we assume by the proportional hazard model that the thing that changes is by slope not affected by time, ie:

$$S(t|x) = [S_0(t)]^{\exp(\beta x)}$$

$\hat{S}_1(t)$ is our estimate for the baseline survival $S_0(t)$, $\beta_2 = 3.13901$ and $\beta_3 = 1.07694$.

We can now calculate:

At $t = 1$:

- Group 2: $\hat{S}_2(1) = [\hat{S}_1(1)]^{HR_2} = (0.9612)^{3.13901} \approx \mathbf{0.8837}$
- Group 3: $\hat{S}_3(1) = [\hat{S}_1(1)]^{HR_3} = (0.9612)^{1.07694} \approx \mathbf{0.9583}$

At $t = 2$:

- Group 2: $\hat{S}_2(2) = [\hat{S}_1(2)]^{HR_2} = (0.9405)^{3.13901} \approx \mathbf{0.8242}$
- Group 3: $\hat{S}_3(2) = [\hat{S}_1(2)]^{HR_3} = (0.9405)^{1.07694} \approx \mathbf{0.9360}$

d.)

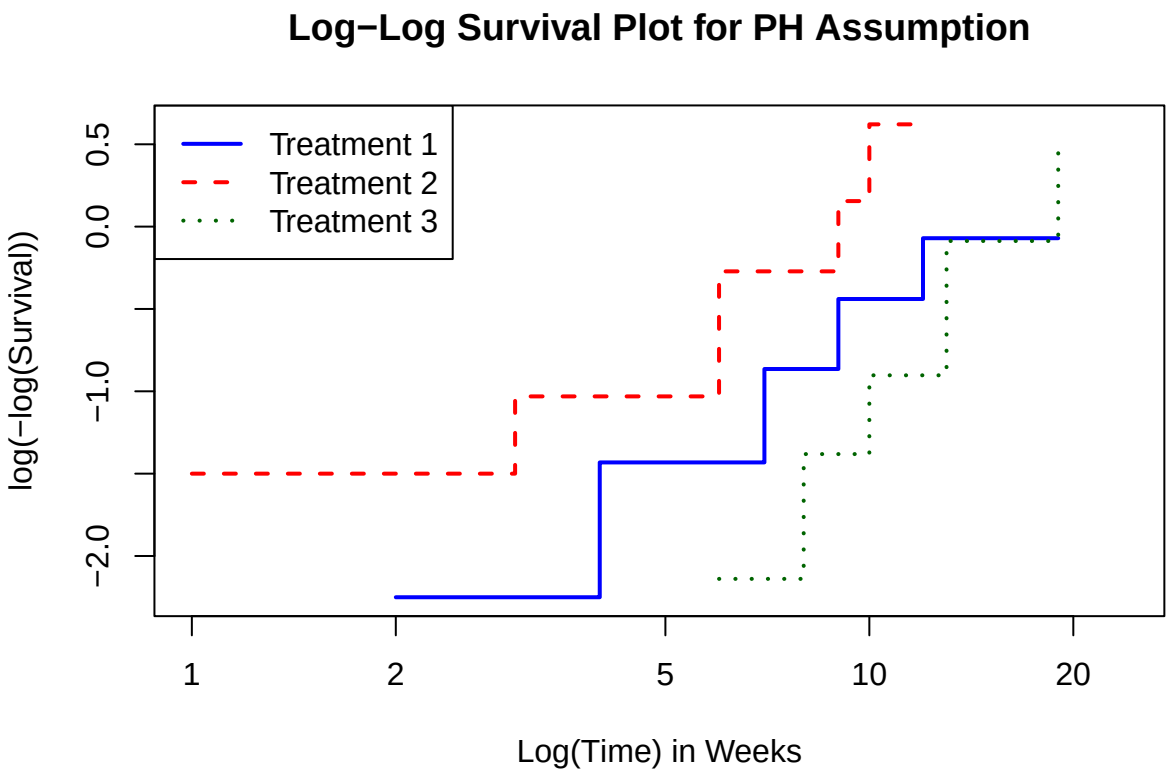
> Provide a graphical assessment to diagnose whether a Cox proportional hazard model can be used

Now we plot the log-log survival plot. We plot log of the $\Lambda(t)$ vs log of the time. This is because the assumption is $log(\Lambda(t)) = log(-logS(t))$ is linear.

```r
# 1. Fit the Kaplan-Meier curves for each treatment group
km_fit <- survfit(Surv(time, cens) ~ trt)

# 2. Create the log-log plot
# note: fun = "cloglog" transforms S(t) to log(-log(S(t)))
plot(km_fit,
     fun = "cloglog",
     col = c("blue", "red", "darkgreen"),
     lty = 1:3,
     lwd = 2,
     main = "Log-Log Survival Plot for PH Assumption",
     xlab = "Log(Time) in Weeks",
     ylab = "log(-log(Survival))")

# 3. Add legend
legend("topleft",
       legend = c("Treatment 1", "Treatment 2", "Treatment 3"),
       col = c("blue", "red", "darkgreen"),
       lty = 1:3,
       lwd = 2)
```



Its hard to say if its all that parallel because the steps are so large but I would say it is roughly decently parallel. The proportional hazard model assumption likely can hold here.

Problem 4

> Consider the survival model characterized by the following piecewise-constant hazard function. where π_1 and π_2 are known constants. Based on right-censored data (x_i, δ_i), i = 1,. . . ,n derive the MLEs of λ_1, λ_2, and λ_3 and their standard errors.

This is our assumption:

$$\Lambda(t) = \begin{cases} \lambda_1 t & 0 \leq t < \pi_1 \\ \lambda_1 \pi_1 + \lambda_2 (t - \pi_1) & \pi_1 \leq t < \pi_2 \\ \lambda_1 \pi_1 + \lambda_2 (\pi_2 - \pi_1) + \lambda_3 (t - \pi_2) & t \geq \pi_2 \end{cases}$$

We aren't given scaling or anything like that so we should assume we are dealing with a model more similar to exponential distribution.

Then the likelihood for one observation i is $L_i = [\lambda(x_i)]^{\delta_i} S(x_i)$. We expand:

$$L_i = [\lambda(t_i)]^{\delta_i} \exp\left(-\int_0^{x_i} \lambda(u) du\right)$$

The log likelihood is simply:

$$\ln L_i = \delta_i \ln \lambda(t_i) - \int_0^{t_i} \lambda(u) du$$

We can further simplify because this is a piecewise constant hazard so we simplify $S(t) = \exp(-\int_0^t \lambda(u) du)$ with the idea that the cumulative hazard $\Lambda(t) = \int_0^t \lambda(u) du$ is the sum of the hazards across the intervals the individual has survived. Ie its piecewise so let t_{ij} be the time spent by individual i in interval j, and d_{ij} be an indicator if individual i died in interval j. Then

$$\int_0^{x_i} \lambda(u) du = \lambda_1 t_{i1} + \lambda_2 t_{i2} + \lambda_3 t_{i3}$$

Now we do this over all the individuals to find the log-likelihood:

$$\ell(\lambda_1, \lambda_2, \lambda_3) = \sum_{i=1}^{n} \sum_{j=1}^{3} (d_{ij} \ln \lambda_j - \lambda_j t_{ij})$$

Let total time at risk in interval j: $T_j = \sum_{i=1}^{n} t_{ij}$ and total events in interval j: $D_j = \sum_{i=1}^{n} d_{ij}$. Importnat to note here that we are counting for time spent inside the interval. If we only know total time alive, we can easily calculate for interval 1: $t_{i1} = \min(t_i, \pi_1)$, for interval 2: $t_{i2} = \max(0, \min(t_i, \pi_2) - \pi_1)$, for interval 3: $t_{i3} = \max(0, t_i - \pi_2)$.

We simplify further, swap the summation order:

$$\ell(\lambda_1, \lambda_2, \lambda_3) = \sum_{j=1}^{3} \left[\left(\sum_{i=1}^{n} d_{ij} \right) \ln \lambda_j - \lambda_j \left(\sum_{i=1}^{n} x_{ij} \right) \right] = \sum_{j=1}^{3} (D_j \ln \lambda_j - \lambda_j T_j)$$

Ok, we have simplified formula. Now, to find the MLE for each λ_j, we take the partial derivative with respect to λ_j and we set it to zero:

$$\frac{\partial \ell}{\partial \lambda_j} = \frac{D_j}{\lambda_j} - T_j = 0 \implies \hat{\lambda}_j = \frac{D_j}{T_j}$$

This is standard MLE of exponential distribution.

Finally, the standard error. The terms are sums of independent terms of each λ_i so they should be independent. We can simply solve for the variance based on Fisher information number:

Second derivative (based on previous calculations):

$$\frac{\partial^2 \ell}{\partial \lambda_j^2} = -\frac{D_j}{\lambda_j^2}$$

Negative inverse:

$$\widehat{Var}(\hat{\lambda}_j) = \left(-\frac{\partial^2 \ell}{\partial \lambda_j^2}\bigg|_{\hat{\lambda}_j} \right)^{-1} = \frac{\hat{\lambda}_j^2}{D_j} = \frac{(D_j/T_j)^2}{D_j} = \frac{D_j}{T_j^2}$$

Finally, the standard error is the square root:

$$SE(\hat{\lambda}_j) = \frac{\sqrt{D_j}}{T_j}$$